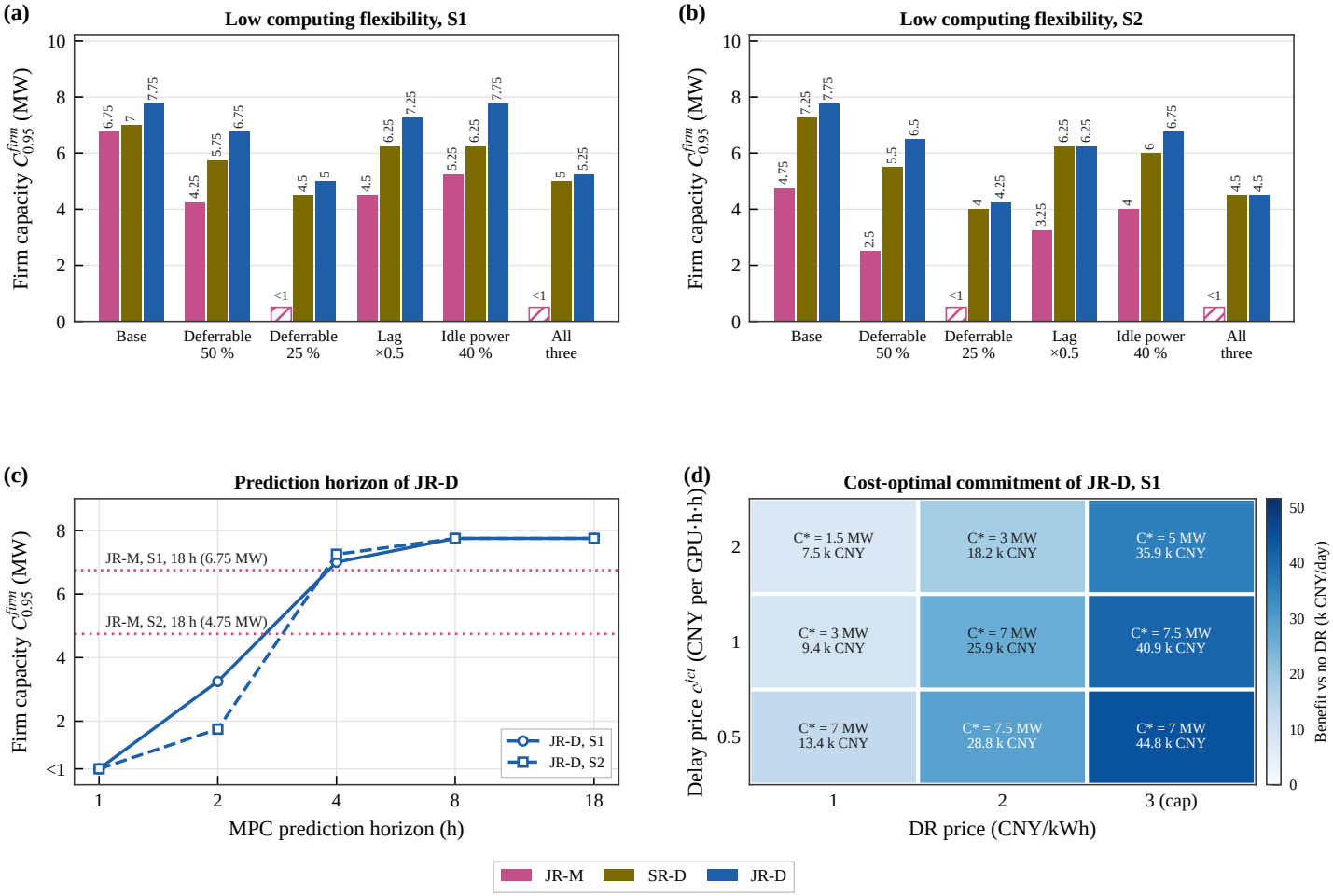
(a)
Low computing flexibility, S1
Firm capacity $C_{0.95}^{firm}$ (MW)
0
2
4
6
8
10
6.75
7
7.75
4.25
5.75
6.75
<1
4.5
5
4.5
6.25
7.25
5.25
6.25
7.75
<1
5
5.25
Base
Deferrable 50 %
Deferrable 25 %
Lag ×0.5
Idle power 40 %
All three
(b)
Low computing flexibility, S2
Firm capacity $C_{0.95}^{firm}$ (MW)
0
2
4
6
8
10
4.75
7.25
7.75
2.5
5.5
6.5
<1
4
4.25
3.25
6.25
6.25
4
6
6.75
<1
4.5
4.5
Base
Deferrable 50 %
Deferrable 25 %
Lag ×0.5
Idle power 40 %
All three
(c)
Prediction horizon of JR-D
Firm capacity $C_{0.95}^{firm}$ (MW)
<1
2
4
6
8
JR-M, S1, 18 h (6.75 MW)
JR-M, S2, 18 h (4.75 MW)
JR-D, S1
JR-D, S2
1
2
4
8
18
MPC prediction horizon (h)
(d)
Cost-optimal commitment of JR-D, S1
Delay price c^{jct} (CNY per GPU·h·h)
2
1
0.5
C* = 1.5 MW
7.5 k CNY
C* = 3 MW
18.2 k CNY
C* = 5 MW
35.9 k CNY
C* = 3 MW
9.4 k CNY
C* = 7 MW
25.9 k CNY
C* = 7.5 MW
40.9 k CNY
C* = 7 MW
13.4 k CNY
C* = 7.5 MW
28.8 k CNY
C* = 7 MW
44.8 k CNY
1
2
3 (cap)
DR price (CNY/kWh)
Benefit vs no DR (k CNY/day)
0
10
20
30
40
50
JR-M
SR-D
JR-D